

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 4 : Session – 8 : SLO – 2 | Micro-programmed Control — Choose the Correct Answer

■ I. Multiple Choice

1. b) Control Store (ROM) — microinstructions reside in a dedicated control store inside the CPU.
2. b) A single control signal — in horizontal microcode each bit drives one control line directly.
3. b) Micro-PC — the micro program counter holds the address of the next microinstruction.
4. b) Shorter and encoded — vertical microcode packs signals into encoded fields, so words are narrow but need decoding.
5. c) Determine next microinstruction address — the microsequencer handles microprogram flow (sequencing/branching).
6. b) Writable Control Store (WCS) — a writable (RAM) control store allows changing microprograms at run time (dynamic microprogramming).
7. a) Smaller control steps within microinstructions — nanocode is a second level below microcode, further breaking microinstructions into nano-steps.
8. b) Complex CISC instructions — microprogramming makes complicated multi-step instructions manageable.
9. b) Easier modification — changing the control unit means editing microcode rather than rewiring hardware.
10. b) Emulation — using microcode to simulate another (complex) instruction set is called emulation.